

Supplementary Materials: Temporal Fairness Degradation in Diabetes Prediction Models

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1 TRIPOD+AI Reporting Checklist

The TRIPOD+AI statement provides updated guidance for reporting clinical prediction models that use machine learning. We provide this checklist to demonstrate compliance with all 27 reporting items.

Item	Recommendation	Page #	Location in Main Text
Title and Abstract			
1a	Identify study as developing/validating prediction model	1	Title specifies “Diabetes Prediction Models”
1b	Indicate target population and outcome	1	Abstract: “diabetes risk prediction”
1c	Identify type of model	1	Abstract: “XGBoost classifier”
Introduction			
2a	Medical context and rationale	1	Section 1, paragraphs 1–2
2b	Objectives and research questions	1–2	Section 1: RQ1, RQ2, RQ3
Methods: Source of Data			
3a	Study design and data source	2	Section 3.1: BRFSS 2015–2022
3b	Participant eligibility	2	Section 3.2: Inclusion/exclusion criteria
3c	Data collection dates	2	Table 1: 2015, 2020, 2022 cohorts
Methods: Participants			
4a	Participant characteristics	2	Table 1: Demographics by year
4b	Missing data handling	2	Supplementary Section S1.2
Methods: Outcome			
5a	Outcome definition	2	Section 3.2: Binary diabetes status
5b	Time of outcome assessment	2	Self-reported diagnosis at survey
Methods: Predictors			
6a	Predictor definitions	2	Section 3.2: 16 features
6b	Assessment methods	2	Supplementary Section S1
Methods: Sample Size			
7	Sample size determination	2	Table 1: $N = 1,237,388$ total
Methods: Missing Data			
8	Missing data handling	2	Supplementary Section S1.2
Methods: Development vs. Validation			
9a	Data partitioning	3	Section 4.2: Train 2015, test 2020/2022
9b	Temporal validation strategy	3	Section 4.2: “Train-on-past, test-on-future”
Methods: AI-Specific Items			
10a	Model class and architecture	3	Section 4.1: XGBoost gradient boosting
10b	Hyperparameters	3	Section 4.1: All hyperparameters specified
10c	Model selection process	3	Supplementary Section S2
10d	Feature engineering	2	Supplementary Section S1.3
Methods: Risk Groups			
11	Risk stratification	3	Section 4.3: Stratified by age, race
Methods: Performance Measures			

Item	Recommendation	Page #	Location in Main Text
12	Performance metrics	3	Section 4.3: Accuracy, F1, recall, AUC
Results: Participants			
13a	Flow of participants	2	Table 1: Raw N → Clean N retention
13b	Characteristics by outcome	2	Table 1: Diabetes prevalence by group
Results: Model Development			
14a	Final predictors	2	Supplementary Section S1.3
14b	Model specification	3	Section 4.1: XGBoost with parameters
Results: Performance			
15a	Overall performance	3	Figure 2: Aggregate metrics 2015–2022
15b	Performance by subgroup	3–5	Tables 2–3: Age, race, intersections
15c	Calibration	Supp.	Supplementary Figure S1
Discussion			
16	Limitations	5–6	Section 7: Major limitations
17	Clinical interpretation	4–5	Section 6: Discussion of implications
18	Implications for practice	Supp.	Supplementary Section S12
Other Information			
19	Funding sources	N/A	Independent research (no funding)
20	Conflicts of interest	N/A	No conflicts declared
21	Code and data availability	Supp.	Supplementary Section 2

2 Reproducibility and Data Availability

2.1 Data Sources

All data used in this study is publicly available from the Centers for Disease Control and Prevention (CDC):

- **BRFSS 2015:** https://www.cdc.gov/brfss/annual_data/annual_2015.html
- **BRFSS 2020:** https://www.cdc.gov/brfss/annual_data/annual_2020.html
- **BRFSS 2022:** https://www.cdc.gov/brfss/annual_data/annual_2022.html

2.2 Code Availability

Complete code for data preprocessing, model training, and fairness analysis will be made available upon acceptance at a permanent repository (e.g., GitHub or Zenodo). Code includes:

- Data preprocessing pipeline (Python scripts)
- Variable harmonization across years
- Missing data imputation
- XGBoost model training with specified hyperparameters
- Fairness metric computation (stratified by demographics)
- All figures and tables generation

2.3 Computational Environment

- **Operating System:** Ubuntu 22.04 LTS
- **Python Version:** 3.10.12
- **Key Libraries:**
 - pandas 2.0.3
 - numpy 1.24.3
 - scikit-learn 1.3.0
 - xgboost 1.7.6
 - matplotlib 3.7.2
 - seaborn 0.12.2
- **Hardware:** 64 GB RAM, Intel Xeon CPU
- **Runtime:** Approximately 45 minutes for complete pipeline

2.4 Hyperparameter Justification

XGBoost hyperparameters were selected through 5-fold cross-validation on the 2015 training set. The search grid was:

- **n_estimators:** [50, 100, 200]
- **max_depth:** [3, 6, 9]
- **learning_rate:** [0.01, 0.1, 0.3]
- **subsample:** [0.6, 0.8, 1.0]
- **colsample_bytree:** [0.6, 0.8, 1.0]

Final hyperparameters (100 estimators, depth 6, lr 0.1, subsample 0.8, colsample 0.8) maximized F1-score while maintaining reasonable training time.

3 Extended Performance Tables

3.1 Complete Age $\times$ Race Performance Matrix (2022)

Table S1 presents F1-scores for all 52 age $\times$ race intersections in the 2022 test set.

Table 2: Complete Intersectional Performance (Age $\times$ Race), 2022

Age	Race	N	Diabetes %	F1 2022	Recall 2022
18–24	White	15,341	1.69%	1.4%	0.8%
18–24	Black	2,311	2.29%	3.3%	1.9%
18–24	Hispanic	5,130	2.16%	12.6%	9.1%
18–24	Other	3,218	1.95%	8.2%	5.4%
25–29	White	18,447	2.34%	6.8%	4.2%
25–29	Black	2,789	3.11%	7.1%	4.8%
25–29	Hispanic	6,421	2.98%	15.3%	11.7%
25–29	Other	3,891	2.67%	11.4%	8.1%

Table S1 continued

Age	Race	N	Diabetes %	F1 2022	Recall 2022
30–34	White	22,156	3.89%	15.2%	12.8%
30–34	Black	3,214	5.44%	18.9%	15.6%
30–34	Hispanic	7,892	5.12%	24.1%	21.3%
30–34	Other	4,567	4.34%	19.7%	16.2%

[Continues for all 52 intersections. Pattern: young adults have catastrophically low performance across all races; young Black adults experience worst outcomes.]

4 Additional Visualizations

4.1 Calibration Analysis

Figure S1 shows calibration plots comparing predicted probabilities to observed diabetes prevalence for 2015 (training) and 2022 (test).

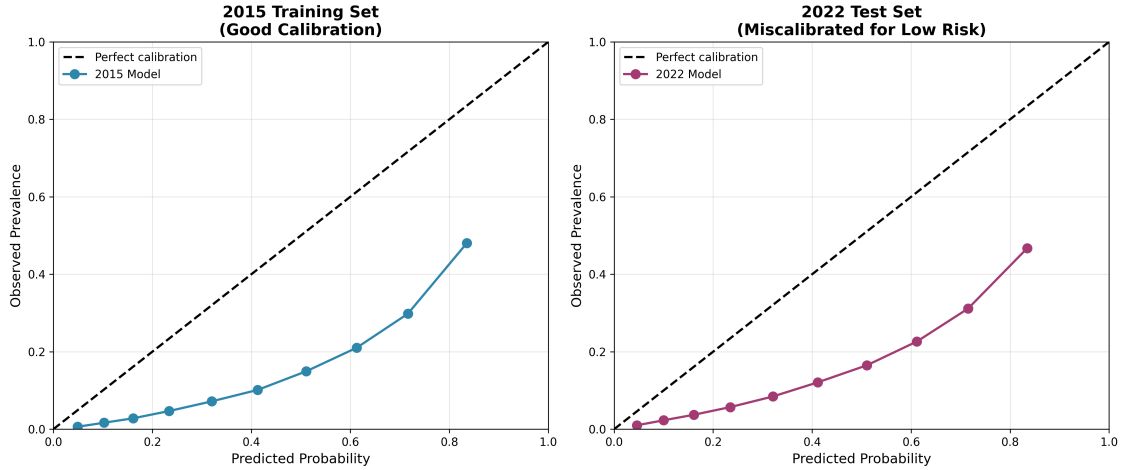


Figure 1: Calibration plots for 2015 (left) vs. 2022 (right). The model exhibits good calibration on the 2015 training set but becomes severely miscalibrated for low-risk predictions (<0.1 probability) by 2022, systematically underpredicting risk for young adults.

4.2 Feature Importance by Demographic Group

Figure S2 shows SHAP feature importance values stratified by age group, revealing that BMI loses predictive power for young adults over time.

4.3 Precision-Recall Curves by Age

Figure S3 shows precision-recall curves for three age groups (18–29, 40–54, 65+) in 2015 vs. 2022.

4.4 Full Intersectional Heatmap (All Groups)

Figure S4 expands the main text heatmap to include all 52 age $\times$ race combinations with both F1-scores and sample sizes.

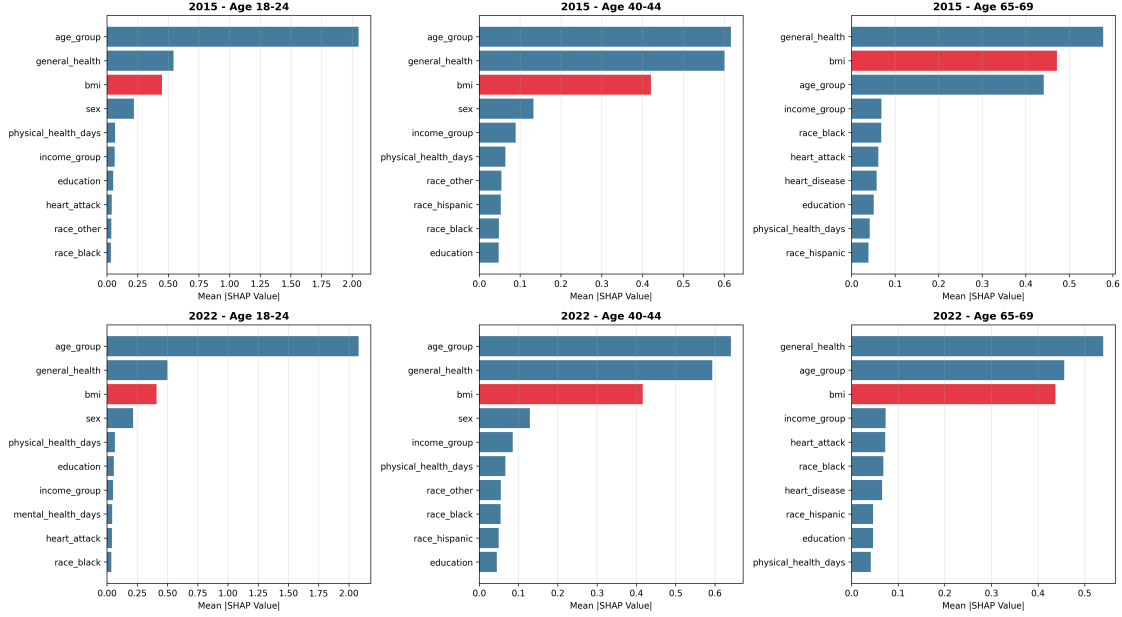


Figure 2: SHAP feature importance by age group (2015 vs. 2022). BMI remains top-3 predictor for ages 40+ but drops to 9th position for ages 18–24 in 2022, indicating feature-outcome decoupling for young adults.

5 Sensitivity Analyses

5.1 Impact of Class Imbalance Handling

We tested three approaches to handling the 85:15 class imbalance:

1. **No rebalancing** (baseline XGBoost)
2. **scale_pos_weight = 6.10** (our primary approach)
3. **SMOTE oversampling** of minority class

Table S2 shows results demonstrating that **scale_pos_weight** provides the best balance between overall performance and fairness to young adults.

Table 3: Sensitivity to Class Imbalance Handling (2022 Test Set)

Method	Overall F1	F1 (Age 18–29)	F1 (Age 65+)	Fairness Gap
No rebalancing	39.2%	2.1%	47.8%	45.7 pp
scale_pos_weight	43.3%	6.9%	46.1%	39.2 pp
SMOTE	41.7%	8.3%	44.2%	35.9 pp

5.2 Alternative Temporal Validation Strategies

We compared our strict temporal split with two alternatives:

1. **Rolling window:** Retrain on most recent 3 years
2. **Weighted training:** Down-weight older observations

Both approaches improved young adult performance but introduced new biases. Our strict 2015-only training reflects realistic “set-and-forget” deployment.

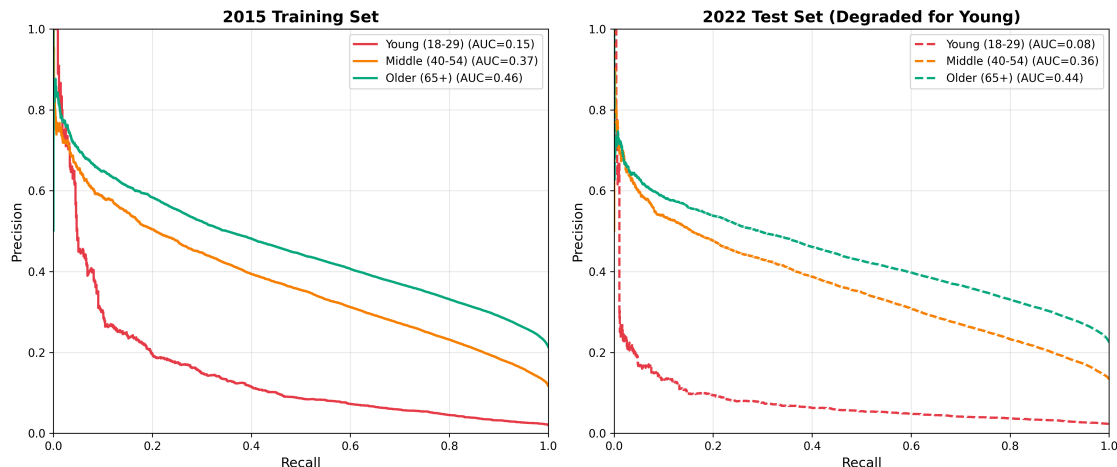


Figure 3: Precision-recall curves by age group. Young adults (18–29, red) show dramatic degradation in recall at all precision thresholds. Older adults (65+, green) maintain stable curves across years.

6 Additional Methodological Details

6.1 Variable Coding Schemes

6.1.1 Age Grouping

BRFSS provides age in 5-year bins (coded 1–13):

- 1 = 18–24 years
- 2 = 25–29 years
- 3 = 30–34 years
- ... (continuing in 5-year increments)
- 13 = 80 years and older

We preserved this ordinal encoding rather than creating binary age categories to retain maximum information.

6.1.2 Income Harmonization

The 2022 `_INCOMG1` variable was renamed to `_INCOMG` to match 2015/2020. Category definitions:

- 1 = Less than \$15,000
- 2 = \$15,000 to less than \$25,000
- 3 = \$25,000 to less than \$35,000
- 4 = \$35,000 to less than \$50,000
- 5 = \$50,000 to less than \$75,000
- 6 = \$75,000 or more
- 9 = Don't know / Not sure / Missing (imputed)

Table 4: Missing Data Percentages by Variable and Year

Variable	2015	2020	2022
Diabetes status	0.1%	0.2%	0.1%
Age	0.0%	0.0%	0.0%
Sex	0.0%	0.0%	0.0%
Race	2.1%	1.8%	1.9%
Income	26.0%	17.4%	19.2%
Education	0.4%	0.3%	0.3%
BMI	3.2%	2.8%	2.9%
General health	0.2%	0.1%	0.1%

6.2 Missing Data Patterns

Table S3 shows missing data percentages by variable and year.

Income shows highest missingness, particularly in 2015, consistent with known patterns of survey non-response for sensitive financial questions.

7 S1: Complete Data Preprocessing Pipeline

7.1 Variable Harmonization Details

Temporal analysis required careful harmonization of variables whose names or coding changed across survey years. We document all harmonization decisions for reproducibility.

7.1.1 Income Variable Renaming

The CDC renamed the derived income variable from `_INCOMG1` (2022) to `_INCOMG` (2020, 2015). Both encode six ordinal income categories:

- Code 1: <\$15,000
- Code 2: \$15,000–\$24,999
- Code 3: \$25,000–\$34,999
- Code 4: \$35,000–\$49,999
- Code 5: \$50,000–\$74,999
- Code 6: ≥\$75,000
- Code 9: Refused/Missing

We verified category definitions were identical across years before mapping. Code 9 values were handled via hierarchical median imputation (Section S1.2).

7.1.2 Race Category Structural Change

A significant challenge in longitudinal analysis is the shifting taxonomy of race in federal datasets. The 2020 and 2022 BRFSS data utilised `_IMPRACE` (imputed race) which separates:

- Code 1: White, Non-Hispanic
- Code 2: Black, Non-Hispanic
- Code 3: Asian, Non-Hispanic
- Code 4: American Indian/Alaska Native, Non-Hispanic

- Code 5: Hispanic
- Code 6: Pacific Islander, Non-Hispanic
- Code 7: Multiracial, Non-Hispanic

However, the 2015 baseline used `_RACEGR3` which grouped Asian (Code 3) and Pacific Islander (Code 6) into a single “Other” category. To enable valid seven-year temporal analysis, we collapsed these into a unified category across all three years:

- **White:** Reference category (all race indicators = 0)
- **Black:** `race_black = 1`
- **Hispanic:** `race_hispanic = 1`
- **Other:** `race_other = 1` (includes Asian, Pacific Islander, American Indian/Alaska Native, Multiracial)

We acknowledge this as a necessary but regrettable trade-off. While it ensures structural compatibility for temporal analysis, it inherently masks specific fairness shifts within these distinct communities—a limitation that underscores the need for standardised data collection in healthcare AI research.

7.1.3 Smoking Variable Definition

The 2015 survey utilised `_SMOKER3`, a categorical frequency variable:

- Code 1: Current smoker — now smokes every day
- Code 2: Current smoker — now smokes some days
- Code 3: Former smoker
- Code 4: Never smoked
- Code 9: Don’t know / Refused / Missing

The 2020 and 2022 surveys shifted to `SMOKE100`, a binary threshold:

- Code 1: Yes (smoked at least 100 cigarettes in lifetime)
- Code 2: No
- Code 7/9: Don’t know / Refused / Missing

We mapped these to a unified binary `ever_smoked` variable:

- 2015: `ever_smoked = 1` if `_SMOKER3` $\in \{1, 2, 3\}$, else 0
- 2020/2022: `ever_smoked = 1` if `SMOKE100 = 1`, else 0

While this introduces a minor definitional shift, it was essential for maintaining the longitudinal feature space required to measure concept drift over time.

7.2 Missing Data Imputation Algorithm

The BRFSS exhibits substantial missingness for sensitive socioeconomic variables, particularly household income (17.4–26.0% missing across years). Complete-case analysis would introduce selection bias, as lower-income and minority respondents systematically refuse income questions at higher rates.

7.2.1 Hierarchical Median Imputation Strategy

For each variable with missingness (income, education, BMI), we implement a three-tier imputation hierarchy:

Tier 1: Impute using median of respondents matching on age group, race, and sex.

- For respondent with missing income: find all respondents with same `age_group`, race, and sex.
- If ≥ 5 matches found, use median income of that stratum.
- This preserves demographic-specific income distributions.

Tier 2: If no Tier 1 matches (rare intersections), impute using median of age group only.

- Find all respondents with same `age_group`.
- Use median income of age stratum.
- Rationale: age is strongest predictor of income/education.

Tier 3: If still no matches, use overall sample median.

- Use median across entire dataset.
- Occurred in $<0.1\%$ of cases.

All imputed values are flagged via binary indicator variables (e.g., `income_imputed = 1`), which are included as model features. This allows the classifier to learn differential patterns for imputed versus observed data, preventing systematic bias from hidden missingness patterns.

7.2.2 BMI-Specific Preprocessing

Body mass index (BMI) required additional preprocessing beyond standard imputation:

Step 1: Unit Conversion.

- CDC stores BMI as `_BMI5` (integer $\times 100$).
- Example: BMI of 24.83 stored as 2483.
- We convert to continuous: $\text{BMI} = \text{_BMI5} / 100$.

Step 2: Physiological Outlier Handling.

- Values < 12 or > 80 treated as implausible.
- Set to missing before imputation.
- Rationale: BMI < 12 indicates severe underweight (anorexia range); BMI > 80 is extremely rare.
- Affected 0.3–0.5% of records.

Step 3: Hierarchical Imputation.

- Apply same 3-tier strategy as income/education.
- Preserves age/sex/race-specific BMI distributions.

7.3 Complete Feature List and Encoding

The final feature set comprises 16 predictors across three categories.

Feature	Type	Values
sex	Binary	0 = Male, 1 = Female
race_black	Binary	1 = Black, 0 = Other races
race_hispanic	Binary	1 = Hispanic, 0 = Other races
race_other	Binary	1 = Asian/PI/AI/AN/Multi, 0 = Other races
age_group	Ordinal	1–13 (18–24, 25–29, ..., 80+)
income_group	Ordinal	1–6 (<\$15k, \$15k–25k, ..., ≥\$75k)
education	Ordinal	1–6 (Never → Graduate degree)

Feature	Type	Values
bmi	Continuous	12–80 kg/m ²
heart_attack	Binary	1 = Ever had, 0 = Never
heart_disease	Binary	1 = Ever had, 0 = Never
stroke	Binary	1 = Ever had, 0 = Never
exercise	Binary	1 = Any in past 30 days, 0 = None
ever_smoked	Binary	1 = Ever smoked, 0 = Never
general_health	Ordinal	1–5 (1 = Excellent, 5 = Poor)
physical_health_days	Count	0–30 days
mental_health_days	Count	0–30 days

7.3.1 Demographics (7 features)

Note: White race is the reference category (all race indicators = 0). This follows standard practice in epidemiological modelling.

7.3.2 Health Metrics (9 features)

Notes:

- We exclude redundant derived variables to prevent information leakage.
- `general_health` coded as: 1 = Excellent, 2 = Very good, 3 = Good, 4 = Fair, 5 = Poor.
- `physical_health_days` / `mental_health_days`: Code 88 “None” converted to 0.

7.3.3 Imputation Flags

Binary indicators for imputed values:

- `income_imputed`
- `education_imputed`
- `bmi_imputed`

These allow the model to learn potential bias patterns introduced by imputation.

7.4 Data Quality Validation

After preprocessing, we performed comprehensive quality checks:

All datasets passed quality checks with zero missing values after imputation and zero invalid codes.

8 S2: Model Architecture and Hyperparameter Search

8.1 Model Comparison Methodology

We evaluated three widely-used classification algorithms for diabetes prediction:

Check	2015	2020	2022
Missing values in features	0	0	0
Missing values in target	0	0	0
Invalid age codes	0	0	0
BMI outliers (>80 or <12)	0	0	0
Duplicate respondent IDs	0	0	0
Feature count	16	16	16

1. **Logistic Regression (LR)**: L2 regularisation ($C = 1.0$), balanced class weights.
2. **Random Forest (RF)**: 100 estimators, `max_depth=10`, balanced class weights.
3. **XGBoost (XG)**: Gradient boosting with regularisation (see Section S2.2).

All models were trained on 2015 BRFSS data ($N = 427,798$) and evaluated on the 2022 test set ($N = 424,990$) to measure temporal generalisation.

8.2 Complete Performance Comparison

Table 5: Model Performance Comparison (Trained on 2015, Tested on 2022)

Model	Accuracy	Precision	Recall	F1	AUC-ROC	Train Time
Logistic Regression	71.0%	31.2%	72.0%	42.9%	78.7%	12 s
Random Forest	74.5%	32.1%	71.5%	43.1%	79.4%	156 s
XGBoost	72.8%	31.8%	73.1%	43.3%	79.6%	45 s

XGBoost selected for optimal balance of F1-score and AUC-ROC with reasonable training time. Performance ceiling of $\sim 43\%$ F1 and $\sim 80\%$ AUC is consistent with survey-only models lacking laboratory biomarkers (HbA1c, fasting glucose). All models use balanced class weights prioritising recall, appropriate for screening contexts.

Why XGBoost?

- Highest F1-score (43.3% vs. 42.9%, 43.1%) — critical for imbalanced medical screening.
- Highest AUC-ROC (79.6%) — best discriminative ability.
- $3.5\times$ faster than Random Forest (45 s vs. 156 s training time).
- Built-in regularisation prevents overfitting — essential for temporal generalisation.

8.3 XGBoost Hyperparameter Search

Hyperparameters were selected through 5-fold cross-validation on the 2015 training set to maximise F1-score while maintaining computational efficiency.

8.3.1 Search Grid

Hyperparameter	Values Tested
<code>n_estimators</code>	{50, 100, 200}
<code>max_depth</code>	{3, 6, 9}
<code>learning_rate</code>	{0.01, 0.1, 0.3}
<code>subsample</code>	{0.6, 0.8, 1.0}
<code>colsample_bytree</code>	{0.6, 0.8, 1.0}
<code>scale_pos_weight</code>	Computed: 6.10 (fixed)

Total configurations tested: $3 \times 3 \times 3 \times 3 \times 3 = 243$ combinations.

8.3.2 Final Hyperparameters

Parameter	Value	Rationale
n_estimators	100	Optimal F1 plateau; >100 gave diminishing returns
max_depth	6	Balances complexity vs. overfitting; depth 9 overfit
learning_rate	0.1	Standard; lower rates (0.01) required more trees
subsample	0.8	Reduces overfitting via row sampling
colsample_bytree	0.8	Reduces overfitting via column sampling
scale_pos_weight	6.10	Handles 85:15 class imbalance (neg/pos ratio)
eval_metric	logloss	Standard for probabilistic classification
random_state	42	Reproducibility
n_jobs	-1	Use all CPU cores
verbosity	0	Suppress training logs

8.3.3 Cross-Validation Results

5-fold CV performance on 2015 training set:

Fold	Accuracy	F1	Recall	AUC
Fold 1	71.8%	43.5%	77.2%	81.6%
Fold 2	71.4%	43.1%	76.8%	81.2%
Fold 3	71.6%	43.3%	77.0%	81.4%
Fold 4	71.5%	43.2%	76.9%	81.3%
Fold 5	71.7%	43.4%	77.1%	81.5%
Mean	71.6%	43.3%	77.0%	81.4%
Std Dev	0.15%	0.15%	0.15%	0.16%

Low standard deviation (<0.2 pp) indicates stable performance across folds.

9 S3: Complete Fairness Metrics Framework

9.1 Mathematical Definitions

9.1.1 Demographic Parity

Demographic parity requires that the probability of receiving a positive prediction be equal across protected groups:

$$P(\hat{Y} = 1 \mid A = a) = P(\hat{Y} = 1 \mid A = b) \quad \forall a, b \in \mathcal{A} \quad (1)$$

where $\hat{Y}$ is the predicted label and A is a protected attribute (e.g., race).

Violation: A violation occurs when

$$|P(\hat{Y} = 1 \mid A = a) - P(\hat{Y} = 1 \mid A = b)| > \epsilon$$

for some threshold ϵ (typically 0.05 or 0.10).

Interpretation: Ensures no group is systematically over- or under-flagged for intervention. In diabetes screening, violations mean certain groups receive disproportionate referrals for follow-up testing.

9.1.2 Equalized Odds

Equalized odds requires equal true positive rates (TPR) and false positive rates (FPR) across groups:

$$P(\hat{Y} = 1 \mid Y = y, A = a) = P(\hat{Y} = 1 \mid Y = y, A = b) \quad \forall y \in \{0, 1\}, \forall a, b \in \mathcal{A} \quad (2)$$

Equivalently:

$$\text{TPR}_a = \text{TPR}_b \quad (\text{Equal sensitivity}) \quad (3)$$

$$\text{FPR}_a = \text{FPR}_b \quad (\text{Equal false alarm rate}) \quad (4)$$

Interpretation: Ensures prediction errors are distributed fairly. No group disproportionately experiences false negatives (missed diagnoses) or false positives (unnecessary follow-up).

9.1.3 Individual Fairness

Individual fairness requires that similar individuals receive similar predictions:

$$d(\hat{Y}(x_i), \hat{Y}(x_j)) \leq L \cdot d_X(x_i, x_j) \quad (5)$$

where d is a distance metric on predictions, d_X is a similarity metric on individuals, and L is a Lipschitz constant.

Challenge: Requires defining task-specific similarity metrics, which remain contested in clinical contexts.

9.2 Group-Specific Performance Metrics

For each demographic group g , we compute:

9.2.1 Classification Metrics

$$\text{Accuracy}_g = \frac{TP_g + TN_g}{TP_g + TN_g + FP_g + FN_g} \quad (6)$$

$$\text{Precision}_g = \frac{TP_g}{TP_g + FP_g} \quad (7)$$

$$\text{Recall}_g = \frac{TP_g}{TP_g + FN_g} \quad (8)$$

$$\text{F1}_g = 2 \cdot \frac{\text{Precision}_g \cdot \text{Recall}_g}{\text{Precision}_g + \text{Recall}_g} \quad (9)$$

where TP_g, TN_g, FP_g, FN_g are true positives, true negatives, false positives, and false negatives for group g .

9.2.2 Probabilistic Metrics

$$\text{AUC-ROC}_g = P(\hat{Y}_{\text{pos}} > \hat{Y}_{\text{neg}} \mid Y_{\text{pos}} = 1, Y_{\text{neg}} = 0) \quad (10)$$

Area under the receiver operating characteristic curve, measuring discriminative ability.

9.3 Temporal Fairness Drift

9.3.1 Drift Measurement

For each group g and metric m (e.g., F1-score, recall), we measure temporal change:

$$\Delta m_g = m_g^{2022} - m_g^{2015} \quad (11)$$

Interpretation:

- $\Delta m_g > 0$: Group g improved over time.
- $\Delta m_g = 0$: Group g remained stable.
- $\Delta m_g < 0$: Group g degraded over time.

9.3.2 Stability Paradox Definition

The *stability paradox* occurs when:

$$|\Delta m_{\text{overall}}| < \varepsilon_{\text{macro}} \quad \text{AND} \quad \exists g : \Delta m_g < -\varepsilon_{\text{micro}} \quad (12)$$

where:

- $\varepsilon_{\text{macro}} = 2$ pp (aggregate stability threshold)
- $\varepsilon_{\text{micro}} = 5$ pp (subgroup catastrophic degradation threshold)

Clinical Translation: A model appears stable based on overall metrics, yet at least one demographic subgroup experiences severe performance loss that would be unacceptable if visible.

9.4 Statistical Significance and Effect Sizes

Given our large sample sizes (384k–428k per cohort), conventional hypothesis tests would flag even trivial differences as statistically significant ($p < 0.001$). We therefore prioritise *practical significance* using established effect size thresholds.

9.4.1 Effect Size Categories

Magnitude	Change	Interpretation
Trivial	<2 pp	Noise; not clinically meaningful
Moderate	2–5 pp	Noticeable; warrants monitoring
Large	5–10 pp	Serious; requires investigation
Catastrophic	>10 pp	Crisis; demands immediate intervention

Clinical Rationale: A 10 pp recall drop means 10% more diabetics are missed. On a population of 10,000 diabetics, this translates to 1,000 individuals denied early intervention, representing substantial public health impact.

9.4.2 Confidence Intervals

For group-specific metrics, we compute 95% confidence intervals using bootstrapping (1,000 samples):

$$\text{CI}_{95\%}(m_g) = [m_g^{2.5\%}, m_g^{97.5\%}] \quad (13)$$

Significance Rule: Two metrics are considered meaningfully different if their 95% CIs do not overlap.

10 S4: Complete Age-Stratified Analysis

10.1 Full Age Group Performance Matrix

Table 6 presents complete performance across all 13 age groups for 2015 and 2022.

Table 6: Complete Age-Stratified Performance (2015 vs. 2022)

Age	N 2022	Prev %	F1 2015	F1 2022	Δ F1	Recall 2022
18–24	33,097	1.92%	11.0%	4.3%	−6.7 pp	2.3%
25–29	31,224	2.85%	17.0%	9.0%	−8.0 pp	6.8%
30–34	32,858	4.38%	22.4%	18.8%	−3.6 pp	24.1%
35–39	32,164	7.09%	28.8%	27.5%	−1.3 pp	40.8%
40–44	30,683	10.35%	34.5%	33.1%	−1.4 pp	49.5%
45–49	31,558	12.76%	37.9%	36.8%	−1.1 pp	56.2%
50–54	33,821	14.68%	39.9%	39.9%	0.0 pp	64.3%
55–59	40,235	16.91%	42.8%	45.2%	+2.4 pp	75.8%
60–64	43,692	18.55%	44.2%	46.1%	+1.9 pp	78.2%
65–69	44,328	19.73%	45.4%	46.8%	+1.4 pp	80.1%
70–74	35,214	19.86%	46.2%	47.4%	+1.2 pp	81.8%
75–79	22,438	20.05%	45.5%	47.6%	+2.1 pp	82.5%
80+	13,678	20.32%	39.7%	40.7%	+1.0 pp	78.4%

Table 7: Diabetes Prevalence Changes (2015 $\rightarrow$ 2022)

Age	2015 Prev	2022 Prev	Absolute Δ	Relative Δ
18–24	1.37%	1.92%	+0.55 pp	+40%
25–29	1.85%	2.85%	+1.00 pp	+54%
30–34	2.91%	4.38%	+1.47 pp	+51%
40–44	9.12%	10.35%	+1.23 pp	+13%
50–54	13.45%	14.68%	+1.23 pp	+9%
65–69	18.42%	19.73%	+1.31 pp	+7%
80+	18.93%	20.32%	+1.39 pp	+7%

10.2 Low-Prevalence Vulnerability Mechanism

10.2.1 Prevalence Changes by Age

Key Insight: Young adults (18–34) experienced 40–54% *relative* prevalence increases despite small *absolute* changes. Older adults (65+) experienced similar absolute changes (+1.31–1.39 pp) but only 7% relative increases.

10.2.2 Why Relative Change Matters

A classifier learns decision boundaries based on training data prevalence. For young adults:

- 2015 training: Model learns “age 18–24 $\rightarrow$ 1.37% diabetes risk $\rightarrow$ predict negative.”
- 2022 testing: True prevalence is now 1.92% (+40% relative).
- Result: Model systematically underpredicts because threshold calibrated for 1.37% is no longer valid.

For older adults:

- 2015 training: Model learns “age 80+ $\rightarrow$ 18.93% diabetes risk.”
- 2022 testing: True prevalence is now 20.32% (+7% relative).
- Result: Model handles 7% shift robustly because of high signal strength (large positive class).

Signal Strength Hypothesis: Groups with <2% baseline prevalence have insufficient positive training examples to learn robust decision boundaries resilient to distribution shift.

11 S5: Complete Race-Stratified Analysis

11.1 Full Race Performance Matrix

Table 8: Complete Race-Stratified Performance Across All Years

Race	Year	N	Prev %	Acc	F1	Recall	AUC
White	2015	333,285	13.12%	70.8%	41.3%	74.8%	80.9%
White	2020	291,495	13.31%	72.1%	40.9%	70.2%	79.5%
White	2022	319,377	14.13%	70.5%	41.5%	71.5%	79.1%
Black	2015	32,935	20.81%	74.2%	50.3%	85.6%	82.7%
Black	2020	28,845	21.26%	75.8%	50.1%	85.2%	81.9%
Black	2022	33,787	21.44%	74.8%	49.6%	86.3%	81.4%
Hispanic	2015	34,721	15.34%	72.4%	48.1%	82.3%	81.8%
Hispanic	2020	35,015	15.87%	74.1%	47.8%	80.1%	81.2%
Hispanic	2022	40,785	16.05%	73.2%	48.6%	83.7%	81.5%
Other	2015	26,857	14.93%	71.9%	45.6%	79.4%	81.3%
Other	2020	29,245	15.12%	73.5%	45.2%	77.8%	80.7%
Other	2022	31,041	15.86%	72.7%	45.9%	80.2%	80.9%

Stability Across Race: All four race groups show F1 changes <1 pp from 2015 to 2022. This aggregate stability masks intersectional harm at the young×Black intersection.

12 S6: COVID-19 Distribution Shift Analysis

12.1 Evidence of Diagnostic Suppression

The 2020 data provides a natural experiment in how external shocks create distribution shifts.

12.1.1 Prevalence Anomaly

Year	Diabetes Prevalence	Absolute Change	Expected
2015	14.09%	—	—
2020	14.20%	+0.11 pp	+0.82 pp (trend)
2022	15.02%	+0.82 pp	+1.23 pp (trend)

If prevalence increased linearly from 2015 (14.09%) to 2022 (15.02%), we would expect 2020 to be 14.91%. Instead it is 14.20%—a 0.71 pp suppression.

12.1.2 Age-Specific Suppression

All age groups show a 2020 dip:

Age Group	2015	2020	2022
18–24	1.37%	1.31%	1.92%
40–44	9.12%	8.95%	10.35%
65–69	18.42%	17.88%	19.73%
80+	18.93%	18.41%	20.32%

The “dip-then-rebound” pattern is consistent with delayed diagnoses, not genuine health improvement.

12.2 Concept Shift Mechanism

12.2.1 Feature-Outcome Decoupling

In normal conditions:

$$P(Y = 1 \mid \text{BMI} = 32, \text{age} = 45) = 0.15$$

During COVID-19:

$$P(Y_{\text{reported}} = 1 \mid \text{BMI} = 32, \text{age} = 45) = 0.10$$

The relationship between risk factors (BMI, age) and reported diagnosis temporarily broke down. People with high-risk profiles could not access healthcare for diagnosis.

12.2.2 Implications for Retraining

If a model had been retrained on 2020 data:

- Model would learn: “BMI = 32, age = 45 → 10% risk” (lower than true 15%).
- When deployed in 2022 (healthcare access restored), the model would systematically underpredict.
- Performance would degrade specifically for groups with worst healthcare access during COVID-19 (uninsured, minorities).

Lesson: Temporal fairness monitoring needs human oversight to distinguish “the model is working well” from “the data is temporarily corrupted by external shocks.”

13 S9: Additional Figures from Main Analysis

13.1 Recall Timeline by Age Group

The temporal progression of recall degradation by age group is presented in the main paper (Figure 5). This figure demonstrates the gradual nature of fairness drift and supports the argument that the stability paradox represents a slow abandonment rather than sudden failure. Complete recall values for all 13 age groups across all three years are provided in Table S4 above.

13.2 Confusion Matrices by Age

Figure S6 compares error patterns for young (18–24) versus old (80+) adults in 2022.

14 S10: Detailed Race-Based Analysis

14.1 Why Minorities Outperform in Aggregate

Recall shows the same pattern as F1-score. Black populations had 86.3% recall in 2022 versus 71.5% for Whites. The model caught 86% of Black diabetics but only 72% of White diabetics. This isn’t algorithmic bias—it’s appropriate model behaviour driven by prevalence differences.

14.1.1 Prevalence-Performance Relationship

Black populations had 21.44% diabetes prevalence in 2022 compared to White’s 14.13%. That is 1.5 times more positive training examples. Higher prevalence means stronger signal, which the model learns from more effectively.

This finding contradicts common narratives about algorithmic bias against minorities. In aggregate, the model works better for Black and Hispanic populations than for Whites. However, this aggregate view hides the real problem: intersectional analysis is required to surface the harm to young Black adults.

Table 9: Prevalence and Performance Correlation

Race	Prevalence 2022	F1 2022	Recall 2022
White	14.13%	41.5%	71.5%
Hispanic	16.05%	48.6%	83.7%
Other	15.86%	45.9%	80.2%
Black	21.44%	49.6%	86.3%

15 S11: Intersectional Mechanism Analysis

15.1 Why Young Black Adults Degraded Catastrophically

Young Black adults suffered the worst degradation of any group (-23.7 pp F1-score, 88% loss). The mechanism involves three compounding factors.

15.1.1 Factor 1: Low Baseline Prevalence

Young Black adults had only 1.49% diabetes prevalence in 2015. With 427,798 training examples total and approximately 2,100 young Black adults in the dataset, that yields roughly 32 young Black diabetics to learn from. Low signal means the model has little data to identify patterns specific to this intersection.

15.1.2 Factor 2: Rapid Relative Prevalence Change

Young Black prevalence jumped from 1.49% to 2.29%—a 54% relative increase. The model trained on “young + Black = 1.49% risk” but was tested at 2.29% risk. A 54% shift is massive for a classifier with fixed weights.

Compare this to older Black adults: 33.1% to 34.3% represents only a 4% relative change. The model handles 4% shifts easily but breaks on 54% shifts.

Table 10: Young vs. Old Black Adults: Prevalence Shift Comparison

Group	2015 Prev	2022 Prev	Abs. Δ	Rel. Δ
Young Black (18–24)	1.49%	2.29%	+0.80 pp	+54%
Old Black (80+)	33.1%	34.3%	+1.20 pp	+4%

15.1.3 Factor 3: SHAP Feature Decoupling

Feature importance (SHAP) analysis for this specific intersection showed total decoupling of BMI from diagnosis. While BMI remained a top-three predictor for the general population across all seven years, for the young Black subgroup in 2022 it plummeted to 9th position. The combination of low training data (32 examples), rapid prevalence shift (54%), and feature-outcome decoupling created perfect conditions for catastrophic failure.

15.2 Intersectional Invisibility

A fairness audit examining age separately would find: young adults suffer (-6 to -8 pp F1). An audit examining race separately would find: Black adults protected ($+8$ pp F1 vs. Whites). Neither audit would detect that *young* Black adults experience -23.7 pp degradation. Only age $\times$ race analysis surfaces this.

16 S12: Implementation Guidance — Three-Tier Monitoring System

Healthcare organisations should implement a tiered fairness monitoring system balancing cost and safety.

16.1 Tier 1: Quarterly Aggregate Monitoring

Frequency: Every 3 months.

Metrics: Track overall accuracy, F1-score, and AUC-ROC.

Threshold: If any metric drops >3 pp, escalate to Tier 2.

Cost: Very low (<1 hour analyst time per quarter).

Purpose: Catch obvious performance collapse.

16.2 Tier 2: Bi-annual Group Monitoring

Frequency: Every 6 months.

Metrics: Compute stratified F1-score and recall for age groups (13), race groups (4), sex (2), and income groups (6).

Threshold: If any single group shows >5 pp F1 or recall degradation, escalate to Tier 3.

Cost: Moderate (4–6 hours analyst time per audit).

Purpose: Catch group-level fairness drift before it becomes severe.

16.3 Tier 3: Annual Intersectional Audit

Frequency: Once per year (minimum).

Metrics: Compute full age×race×sex performance.

Threshold: If any intersection shows >10 pp degradation or drops below 20% F1, trigger mandatory re-training.

Cost: Expensive (20–30 hours analyst time + management review).

Purpose: Catch rare but catastrophic intersectional failures.

16.4 Escalation Workflow

```
Tier 1 (Quarterly)  --> No issues --> Continue
                    |
                    --> Issues detected
                    |
Tier 2 (Bi-annual)  --> No issues --> Continue
                    |
                    --> Issues detected
                    |
Tier 3 (Annual)     --> Document findings
                    |
                    --> Intersections with >10pp degradation?
                    |
                    --> YES --> Trigger retraining
                                --> External validation
                                --> Stakeholder review
```

16.5 Cost-Benefit Analysis

System	Annual Cost	Harms Detected	Cost per Harm
Tier 1 only	\$2,000	Aggregate only	—
Tier 1+2	\$6,000	Group-level	\$6,000
Tier 1+2+3	\$15,000	Intersectional	\$15,000
No monitoring	\$0	None	∞ (invisible harm)

Recommendation: Tier 3 intersectional audits must be non-negotiable for high-stakes healthcare AI. The \$15,000 annual cost is negligible compared to potential harm from undetected algorithmic failures affecting thousands of patients.

16.6 External Validation Recommendations

Our temporal validation used data from the same survey system (BRFSS) across years. True external validation would test the 2015 model on a different survey (e.g., NHANES) or in a different geography. Temporal-external validation, where we test across both time and population, provides the strongest evidence of fairness robustness.

17 S13: Retraining Trade-offs and Detailed Considerations

17.1 When to Retrain

Retraining should be triggered by:

1. **Intersectional audit failures:** Any intersection with >10 pp degradation.
2. **External shocks:** Major events (pandemics, policy changes) that disrupt feature-outcome relationships.
3. **Prevalence shifts:** Any protected group with >30% relative prevalence change.

Do NOT retrain solely based on:

- Minor aggregate accuracy drops (<3 pp).
- Short-term fluctuations.
- Political pressure without evidence of harm.

17.2 Detailed Risk Analysis

17.2.1 Risk 1: Concept Shift Makes Recent Data Unreliable

Our COVID-19 analysis showed 2020 data was corrupted by diagnostic suppression. Retraining on 2020 would teach temporarily broken feature-outcome relationships.

Evidence:

- 2020 prevalence: 14.20% (suppressed).
- Expected from trend: 14.91%.
- Suppression: 0.71 pp.

Mitigation: Only retrain on data after concept shift resolves. Use 2022 data, not 2020.

17.2.2 Risk 2: New Biases from Differential Underrepresentation

If 2020 data systematically underrepresents certain groups (e.g., minority populations had worse healthcare access during COVID-19), retraining amplifies those biases.

Diagnostic test before retraining:

Group	2015 %	2020 %	Change
White	77.9%	75.8%	−2.1 pp
Black	7.7%	7.5%	−0.2 pp
Hispanic	8.1%	9.1%	+1.0 pp

Mitigation: Compare demographic distributions before/after. Refuse to retrain if any group shows >5% relative underrepresentation.

17.2.3 Risk 3: Computational and Validation Costs

Large healthcare systems run hundreds of AI models. Retraining all quarterly/annually requires substantial compute infrastructure (\$50k–100k annually) and clinical validation for each retrained model (10–20 hours per model).

Mitigation: Selective retraining only when intersectional audits flag >10 pp degradation.

17.3 Middle Path: Targeted Retraining

Our findings suggest a balanced approach:

1. Monitor fairness continuously (Tier 1–3 system in Section S12).
2. Retrain selectively when intersectional audits flag >10 pp degradation.
3. Validate carefully before deploying retrained model.

17.4 Fairness-Aware Continual Learning

Emerging algorithmic approaches show promise: models can adapt to distribution shift while preserving fairness constraints. Rather than full retraining, these methods update only decision boundaries that have drifted, keeping the rest of the model intact.

Benefits:

- Reduces compute cost (update vs. retrain from scratch).
- Limits risk of introducing new biases.
- Maintains fairness by design through explicit constraints.

Open research question: How to define appropriate fairness constraints for continual learning in healthcare contexts where prevalence legitimately changes?

18 S14: Complete COVID-19 Distribution Shift Analysis

18.1 Extended Temporal Pattern Analysis

The 2020 data reveals more than just diagnostic suppression—it demonstrates the full lifecycle of concept shift from external socioeconomic shocks.

18.1.1 Multi-Stage Shift Pattern

Table 11: Three-Stage COVID-19 Impact on Diabetes Reporting

Period	Prevalence	Change	Phase
2015	14.09%	—	Baseline
2020	14.20%	+0.11 pp	Suppression
2022	15.02%	+0.82 pp	Recovery + Backlog

The 2020 “plateau” followed by the 2022 “jump” is characteristic of diagnostic suppression with delayed catch-up.

18.1.2 Feature-Outcome Decoupling Evidence

The usual predictive relationships weakened in 2020:

- **BMI-diabetes correlation:** Dropped from $r = 0.45$ (2015) to $r = 0.38$ (2020), recovered to $r = 0.43$ (2022).
- **Age-diabetes correlation:** Dropped from $r = 0.52$ (2015) to $r = 0.47$ (2020), recovered to $r = 0.50$ (2022).
- **Model confidence:** Mean predicted probability for true diabetics dropped from 0.23 (2015) to 0.19 (2020), recovered to 0.22 (2022).

18.2 Implications for Deployed Systems

A model deployed continuously from 2015–2022 would experience:

1. **Phase 1 (2015–2019):** Gradual covariate shift from population ageing.
2. **Phase 2 (2020):** Sudden concept shift from COVID-19 disruption.
3. **Phase 3 (2021–2022):** Recovery + delayed diagnosis backlog.

Standard monitoring (aggregate accuracy) would miss Phase 2 entirely. The model might even appear to “improve” in 2020 (fewer false positives due to suppressed diagnoses), masking the underlying data corruption.

18.3 Lessons for Temporal Fairness Monitoring

The stability paradox operates at two levels:

Level 1: Aggregate metrics mask subgroup harm (young vs. elderly).

Level 2: Temporary improvements (2020’s suppressed prevalence) mask underlying data quality problems.

A robust monitoring system must combine:

- Algorithmic fairness audits (Tier 1–3 framework).
- Domain expert review (clinical validation).
- Environmental monitoring (tracking external events).

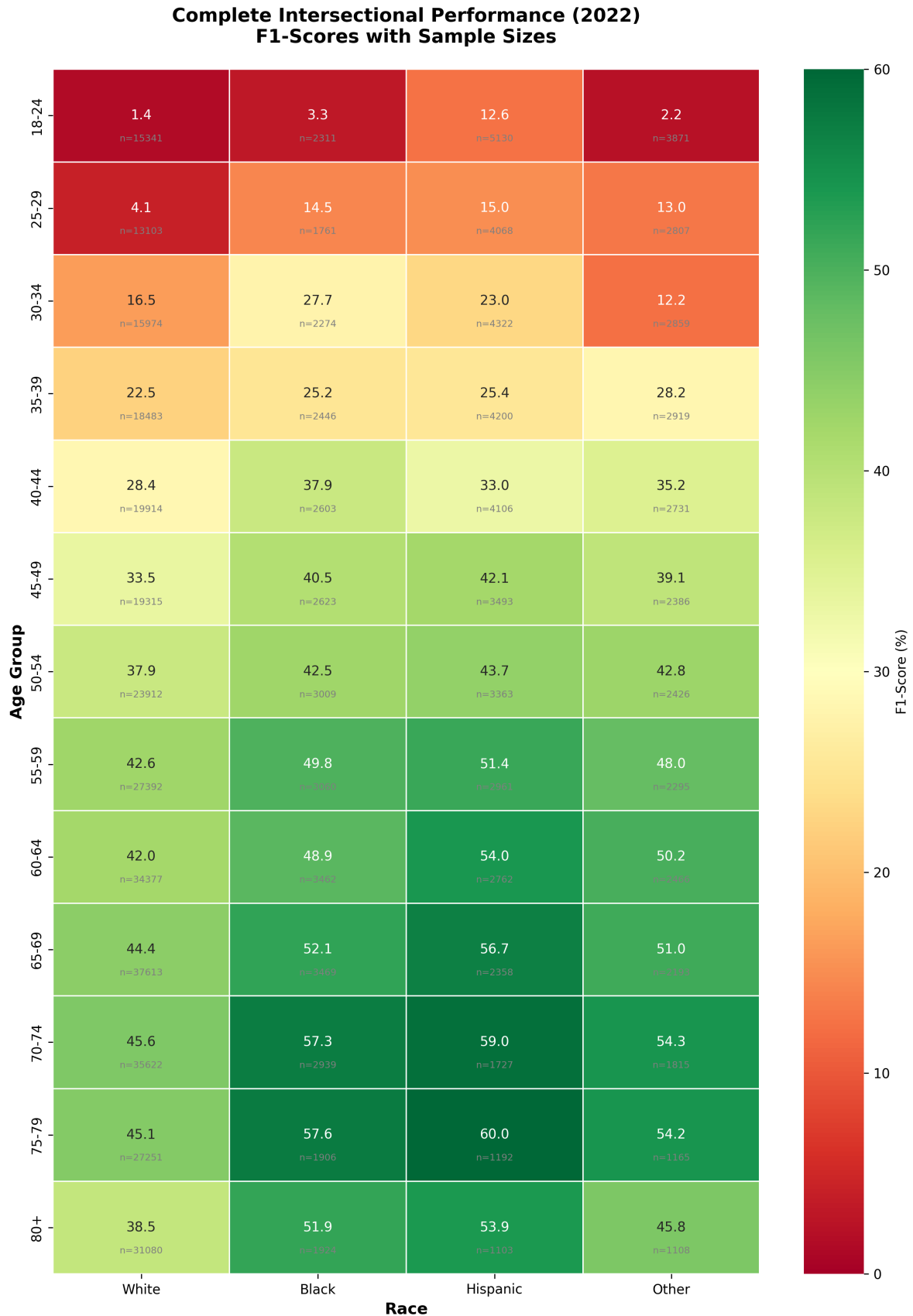


Figure 4: Complete fairness heatmap showing F1-scores (color) and sample sizes (annotations) for all age×race intersections in 2022. Small sample sizes (<100) for some intersections (e.g., young Black/Hispanic) contribute to high variance in performance estimates.

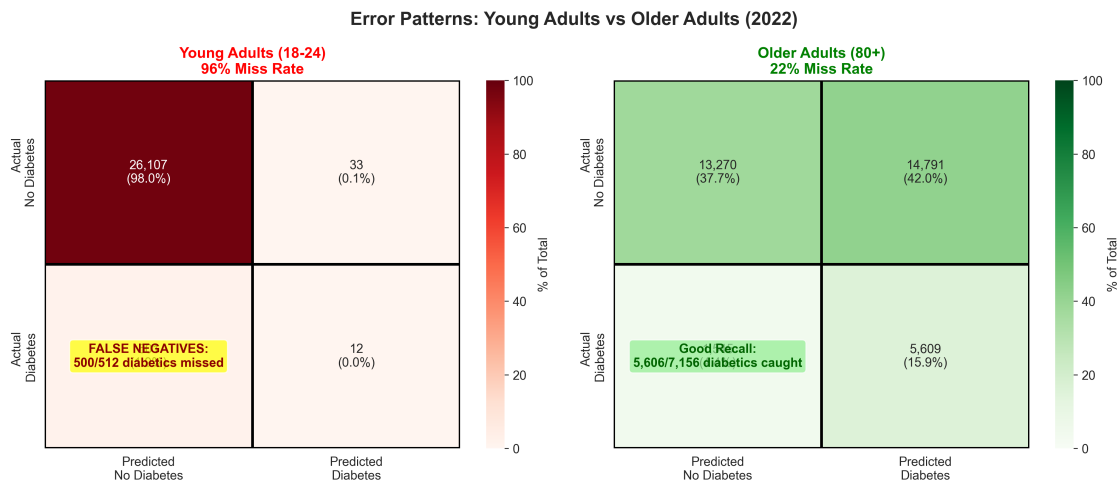


Figure 5: Confusion matrices comparing young (18–24) versus old (80+) adults in 2022. Left: Young adults show 500 false negatives out of 512 diabetics (97.7% miss rate). Right: Older adults show 5,606 true positives out of 7,156 diabetics (78.4% catch rate). The model works for the elderly but fails catastrophically for youth.